Discrete Math Class Notes

These are class notes for a first year undergrad Discrete Math class taught from Grimaldi's 'Discrete and Combinatorial Mathematics (5th edition)', which we will refer to as 'the text'.

1 Counting

This section draws from Sections 1.1 - 1.4 of the text. It is understood that the students have seen basic counting techniques, so we only review these topics quickly. Many of what would be examples are posed as problems.

Topics Covered:

- Rule of Sum
- Rule of Product
- Counting the Complement
- Permutations of sets and multisets
- Circular permutations
- Combinations
- Combinations with replacement

A *set* is a collection of objects, which we call *elements* or *members* of the set. For the time being, the sets we consider will be finite sets in which no member occurs more than once.

Example 1.1. The collection S of two digit integers not divisible by 3 is a set:

$$S = \{10, 11, 13, 14, 16, 17, 19, \ldots, 97, 98\}$$

We are interested in counting $|S|$, the number of elements in S. Clearly we could just count, and find that there are 60. There are some basic techniques that allow us to simplify this task.

1.1 Basic Counting Techniques:

The Rule of Sum (Divide and Conquer): If we can break a set A into smaller sets $A_1, \ldots, A_d$ such that every element is in exactly one set, then

$$|A| = |A_1| + \cdots + |A_d|.$$

For example, we could break the set S from Example 1.1 up into the set $S_1 = \{10, 13, 16, 19, \ldots, 94, 97\}$ of the form $3x + 1$ for some integer x, and the set $S_2 = \{11, 14, 17, 20, \ldots, 95, 98\}$ of the form $3x + 2$. It is not hard to see that $|S_1| = |S_2|$ so we only have to count one of them to see that $|S| = |S_1| + |S_2| = 30 + 30 = 60$.

The Rule of Product: One good way of counting elements of a set is considering the process of choosing an element of the set. If the process can be broken down into independent steps, we can simply cound the ways of completing each step.

For example, the elements of S are integers of the form $3x + y$ where x is in $\{3, 4, \ldots, 32\}$ and y is in $\{1, 2\}$. So we construct every element by choosing an x and a y. We can choose x in 30 ways and y in 2 ways, so there are $30 \cdot 2 = 60$ ways of choosing an element of S.

Counting the Complement: Sometimes it is easy to count a bigger set and then remove items not in the wanted set.

For example, it is easier to count the set T of integers between 1 and 99 that are not divisible by 3. There are 66 of them. The *complement* $\overline{S}$ of S in T, that is the set of elements of T not in S is $\overline{S} = \{1, 2, 4, 5, 7, 8\}$, so $|S| = |T| - |\overline{S}| = 66 - 6 = 60$.

1.2 Permutations and Combinations

1.2.1 Permutations of a set

A *permutation* of a set is an arrangement of the items in a line. So $1, 2, 3$ and $3, 1, 2$ are different permutations of the set $\{1, 2, 3\}$. Using the rule of product, one can show that there are

$$n! = n \cdot (n - 1) \cdot \ldots \cdot 1$$

permutations of an n element set. Indeed, there are n ways to choose the first element, and then $n - 1$ ways to choose the second, etc. The notation '$n!$' is read 'n factorial'.

An r-*permutation* or a *size-r permutation* of set is an arrangement of r items of the set in a line. Again by the rule of product, one can show that there are

$$P(n,r) := \frac{n!}{(n-r)!}$$

r-permutations of an n element set.

1.2.2 Permutations of a multiset

A *multiset* is a collection of elements that are not necessarily distinct. The multiset $\{1,2,2,2\}$ is different from $\{1,2\}$, though as sets, they would be the same. Counting permutations of a multiset adds a slight complication to the task of counting permutations of a set: certain permutations are indistinguishible, so we consider them the same. The multiset $\{1,2,2,2\}$ only has 4 distinguishible permutations:

$$\{(1,2,2,2),(2,1,2,2),(2,2,1,2),(2,2,2,1)\}.$$

This is less than the $4! = 24$ permutations of a 4-element set.

We could get this by counting the elements of the multiset as distinct, in $4!$ ways, and then observing that each distinguishible permutation of the multiset is counted $3!$ times, one for each permutation of the repeated elements. So there are $4 = 4!/3!$ distinguisible permutations of the multiset. More generally we have...

Rule: In a multiset containing n_i *indistinguisible* elements of type i for $i = 1,\ldots,r$, the number of *distinguishible* permutations is

$$\frac{(n_1+n_2+\cdots+n_r)!}{n_1!\cdot n_2!\cdot\cdots\cdot n_r!}.$$

When we talk of permutations of a multiset, it is understood that we are talking of distinguishible permutations, so we will generally drop the 'distinguishible.'

Problem 1.1. Find the number of permutations of the letters MISSISSIPPI.

Problem 1.2. Find the number of permutations of the letters ABCDEFGH in which the letters of the word CAGE occur in this order. In which the word CAGE doesn't occur.

Problem 1.3. Count the number of paths in the plane from $(0,0)$ to $(4,9)$, using only steps that go one unit up or 1 unit to the left

Hint

Text Example 1.14

A *circular permutation* of a set is an arrangement of the elements in a circle. Two such arrangements are considered the same if you can get from one to the other by rotating the circle. These are the permutations we are considering in the next problem.

Problem 1.4.

(a) How many ways can 6 people be seated around a circular table. (That is, how many circular permutations are there of a 6 element set?)

(b) How many ways can 3 men and 3 women be seated around a circular table so that men and women alternate.

(c) How many ways can 3 couples be seated around a table so that the couples are seated together.

1.2.3 Combinations without replacement

An r-combination of an n-element set is a choice of r elements from the set. If the r-elements must be distinct, there are

$$\binom{n}{r} = P(n,r)/r! = \frac{n!}{(n-r)!r!}$$

of them, (They are permutations in which order is disregarded.) The notation $\binom{n}{r}$ is read 'n choose r', and is called a *binomial co-efficient* because of the following theorem.

Theorem 1.2 (Binomial Theorem)**.**

$$(x+y)^n = \sum_{i=0}^{n} \binom{n}{i} x^{n-i} y^i$$

Proof. Indeed, $(x+y)^n$ is the product of n factors $(x+y)$. Every summand in the expansion of $(x+y)^n$ corresponds to a choice of either x or y from each of these terms. The summand $x^{n-i}y^i$ will occur once for every way we can choose i of the n terms to choose y from. This is $\binom{n}{i}$ ways. □

Here is the typical problem.

Example 1.3. A pizza place has 16 different toppings. On a deluxe pizza you can choose 6 different toppings. How many different deluxe pizzas are there?

Solution

This is pretty straight forward: $\binom{16}{6}$.

See if you can deal with the following complications of this problem.

Problem 1.5. A pizza place offers the following toppings: 4 kinds of cheese, 7 kinds of vegetables, and 5 kinds of meat. How many different pizzas have

i. 2 cheeses, 4 vegetables and 3 meats (all different).

ii. 6 different toppings at least 3 of which are meat.

iii. 6 different toppings at least 3 of which are cheese.

Some problems can be dealt with as permutations or combination problems.

Example 1.4. How many ways can we split 20 people into 4 teams A, B, C, D of 5 people each. (Teams are distinguishable.)

Solution

We can view this as a permutation problem or a combination problem. We can choose 5 for the first team, 5 for the second, etc., so there are

$$\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}.$$

Alternately, we can arrange 5 of each of A, B, C, D among 20 people:

$$\frac{20!}{5!5!5!5!}$$

You can show that these quantities are the same.

Example 1.5. How many ways can you arrange the letters MISSISSIPPI such that there are no adjacent I's.

Solution

We can do this using combinations and permutations together. First arrange the letters MSSSSPP in $7!/(4!2!)$ ways. Given such an arrangement: SMSPSSP, there are 8 spaces in which we can put one I: -S-M-S-P-S-S-P-. There are $\binom{8}{3}$ ways to place the 3 I's in these spaces, so the total number of arrangements is

$$\frac{7!}{4!2!}\binom{8}{3}.$$

Problem 1.6. What is the co-efficient of x^3y^3 in

i. $(x+y)^6$?

ii. $(2x+y)^6$?

1.3 Combinations with replacement

Consider the following question:

Example 1.6. A doughnut store has kruellers, fritters and honey glazed doughnuts. How many ways can you select a dozen?

Observe that there is a one-to-one correspondence between the ways to select a dozen, and the arrangements of 12 stars: '$*$' and 2 dividers: '|'. The arrangement

$$**\,|\,*\,*\,*\,*\,*\,*\,*\,|\,*\,*\,*$$

partitions the 12 stars into three parts. This corresponds to the choice of 2 kruellers, 7 fritters, and 3 honey glazed. Thus there are $\binom{12+2}{2}$ choices of the dozen. It is clear that this also counts the solutions of

$$x_1 + x_2 + x_3 = 12$$

where x_1, x_2, x_3 are non-negative integers. The value of x_i in a solution corresponds to the number of '$*$'s in the ith part.

Here are a couple of variations on the question.

Problem 1.7. How many solutions are there to the equation $x_1 + x_2 + x_3 = 12$ where all x_i are non-negative integers and $x_2 \geq 2$?

Hint

(1.7) One can do this by counting the non-negative integer solutions of $x_1 + x_2' + x_3 = 10$ (where $x_2' = x_2 - 2$).

Problem 1.8. How many solutions are there to the equation $x_1 + x_2 + x_3 \leq 12$ where all x_i are non-negative integers.

Problem 1.9. A composition of an integer n is an ordered sum of positive integers totalling to n. (So 8, $7+1$, $1+7$ and $1+1+1+1+1+1+1+1$ are all different compositions of 8.) How many compositions are there of 8?

Hint

(1.8) One can do this by counting the non-negative integer solutions of $x_1 + x_2 + x_3 + x_4 = 12$, (where x_4 is called a slack variable.)

Problem 1.10. How many ways can seven 'H's and nine 'T's be arranged in a line so that there exists exactly four strings of consecutive 'T's?